

Week 3: Seller and Inventory Management System

Scenario

The online marketplace works with multiple sellers who provide different products. The system stores seller information, manages supplied products, and keeps track of stock availability.

Tables

- Seller
- Inventory

```
SQL> CREATE TABLE Seller (seller_id VARCHAR2(10) PRIMARY KEY, seller_name VARCHAR2(50) NOT NULL, email VARCHAR2(60) UNIQUE NOT NULL, phone_number VARCHAR2(15) UNIQUE, city VARCHAR2(30), rating NUMBER(2,1) CHECK (rating BETWEEN 1 AND 5));
```

Table created.

```
SQL> CREATE TABLE Inventory (inventory_id VARCHAR2(10) PRIMARY KEY, seller_id VARCHAR2(10), product_name VARCHAR2(50) NOT NULL, stock_quantity NUMBER(5) CHECK (stock_quantity >= 10), unit_price NUMBER(8,2) CHECK (unit_price > 50), warehouse_location VARCHAR2(40), FOREIGN KEY (seller_id) REFERENCES Seller(seller_id));
```

Table created.

Relationships

Seller and Inventory

- One seller can supply many products.
- Each inventory record belongs to one seller.

Relationship:

- Seller → Inventory (One-to-Many)

Constraints

- Primary Key for Seller Table
- Primary Key for Inventory Table
- Foreign Key in Inventory Table
- Email must be UNIQUE
- Email must be NOT NULL
- Phone Number must be UNIQUE
- Rating should be between 1 and 5
- Stock Quantity should be greater than or equal to 10
- Unit Price should be greater than 50

Insert Values

- Insert values into Seller Table

```
SQL> INSERT INTO Seller VALUES ('S201','Apex Traders','apex@gmail.com','9876543210','Chennai',4.5);
1 row created.

SQL> INSERT INTO Seller VALUES ('S202','Green Mart','green@gmail.com','9876543211','Coimbatore',4.2);
1 row created.

SQL> INSERT INTO Seller VALUES ('S203','Star Supplies','star@gmail.com','9876543212','Madurai',4.8);
1 row created.

SQL> INSERT INTO Seller VALUES ('S204','Elite Sellers','elite@gmail.com','9876543213','Trichy',3.9);
1 row created.

SQL> INSERT INTO Seller VALUES ('S205','Future Store','future@gmail.com','9876543214','Salem',4.6);
1 row created.
```

```
SQL> SELECT * FROM Seller;

SELLER_ID  SELLER_NAME
-----
EMAIL
-----
CITY
-----
RATING
-----
S201      Apex Traders
apex@gmail.com
Chennai
4.5
S202      Green Mart
green@gmail.com
Coimbatore
4.2
SELLER_ID  SELLER_NAME
-----
EMAIL
-----
CITY
-----
RATING
-----
S203      Star Supplies
star@gmail.com
Madurai
4.8
S204      Elite Sellers
elite@gmail.com
9876543213
SELLER_ID  SELLER_NAME
-----
EMAIL
-----
CITY
-----
RATING
-----
Trichy
3.9
S205      Future Store
future@gmail.com
Salem
4.6
```

- Insert values into Inventory Table

```
SQL> INSERT INTO Inventory VALUES ('I301','S201','Wireless Mouse',120,650,'Warehouse A')
1 row created.

SQL> INSERT INTO Inventory VALUES('I302','S202','Keyboard',80,850,'Warehouse B');
1 row created.

SQL> INSERT INTO Inventory VALUES('I303','S203','Monitor',45,12500,'Warehouse C');
1 row created.

SQL> INSERT INTO Inventory VALUES('I304','S204','USB Drive',200,450,'Warehouse D');
1 row created.

SQL> INSERT INTO Inventory VALUES('I305','S205','Laptop Stand',60,1200,'Warehouse E');
1 row created.
```

-

```
SQL> SELECT * FROM Inventory;

INVENTORY_ SELLER_ID  PRODUCT_NAME
-----
STOCK_QUANTITY UNIT_PRICE  WAREHOUSE_LOCATION
-----
I301        S201        Wireless Mouse
          120          650 Warehouse A

I302        S202        Keyboard
          80          850 Warehouse B

I303        S203        Monitor
          45         12500 Warehouse C

INVENTORY_ SELLER_ID  PRODUCT_NAME
-----
STOCK_QUANTITY UNIT_PRICE  WAREHOUSE_LOCATION
-----
I304        S204        USB Drive
          200          450 Warehouse D

I305        S205        Laptop Stand
          60          1200 Warehouse E
```

-

Report 1: Seller-wise Product Report

Display:

- Seller Name

- Product Name
- Stock Quantity

```

SQL> SELECT s.seller_name,      i.product_name,      i.stock_quantity FROM Seller s JOIN Inventory i ON s.seller_id = i.seller_id;

SELLER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
Apex Traders
Wireless Mouse                                120

Green Mart
Keyboard                                      80

Star Supplies
Monitor                                       45

SELLER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
Elite Sellers
USB Drive                                    200

Future Store
Laptop Stand                                60

```

Report 2: Stock Availability Report

Display:

- Product Name
- Stock Quantity
- Stock Status

```

SQL> SELECT product_name,      stock_quantity,      CASE WHEN stock_quantity >= 100 THEN 'High Stock' WHEN stock_quantity BETWEEN 50 AND 99 T
HEN 'Medium Stock'      ELSE 'Low Stock'      END AS stock_status FROM Inventory;

PRODUCT_NAME                                STOCK_QUANTITY STOCK_STATUS
-----
Wireless Mouse                                120 High Stock
Keyboard                                      80 Medium Stock
Monitor                                       45 Low Stock
USB Drive                                    200 High Stock
Laptop Stand                                60 Medium Stock

```

Report 3: Inventory Analysis

Find:

- Total Products Available

```

SQL> SELECT COUNT(*) AS total_products FROM Inventory;

TOTAL_PRODUCTS
-----
5

```

- Low Stock Products

```
SQL> SELECT COUNT(*) AS low_stock_products FROM Inventory WHERE stock_quantity < 50;
```

```
LOW_STOCK_PRODUCTS
-----
1
```

-
- Highest Stocked Product

```
SQL> SELECT product_name, stock_quantity FROM Inventory WHERE stock_quantity = (SELECT MAX(stock_quantity) FROM Inventory);
```

```
PRODUCT_NAME                                STOCK_QUANTITY
-----
USB Drive                                200
```

-
- Average Inventory Quantity

```
SQL> SELECT AVG(stock_quantity) AS average_inventory FROM Inventory;
```

```
AVERAGE_INVENTORY
-----
101
```

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This follows the same structure as the PDF.